

20XW86 – INFORMATION RETRIEVAL AND WEB SEARCH LAB

NEWSHIVE – SMART NEWS AGGREGATOR BOT

TEAM MEMBERS:

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ONE-LINER:

A Telegram Bot that delivers personalized, relevant news based on semantic search, media trends, and Google Sheets export.

A news aggregator that fetches news from NewsAPI, summarizes the results, and delivers them via a Telegram bot. The results are also written to a Google Spreadsheet for archival and further analysis.

PURPOSE

This project is designed to streamline news retrieval by allowing users to fetch articles from specific sources using keywords and semantic search. The Telegram bot provides a brief summary of articles, helping users decide which ones to read, thereby reducing time spent browsing multiple websites. The system ensures personalized and efficient news delivery with real-time configurability.

FEATURES

- Fetch news from NewsAPI based on user-defined topics.
- Summarize articles for quick review.
- Deliver results via a Telegram bot.
- Store fetched articles in a Google Spreadsheet for real time configuration.
- Customize news sources and search timeframes.
- Integrate relevant media (images/videos).
- Semantic Search using SentenceTransformer embeddings.
- Hybrid Ranking: Combine keyword relevance and vector similarity.
- Media Filtering: Retrieve related images (Pixabay) and videos (YouTube).
- Telegram Integration for interactive delivery.

API'S USED

- **NewsAPI** – For retrieving news articles.
- **Telegram Bot API** – For user interaction and message delivery.
- **Google Sheets API** – For storing and exporting article data.
- **Pixabay API** – For retrieving images.
- **YouTube API** – For video embedding.

TECHNOLOGY STACK:

- **Languages & Frameworks:** Python
- **Libraries:** SentenceTransformers, Requests, Pandas, gspread
- **Database:** ChromaDB (Vector Store for semantic search)

METHODOLOGY / APPROACH:

- Fetched articles using NewsAPI with user keywords or configurations.
- Pre-processed content and generated SentenceTransformer embeddings.
- Stored embeddings in ChromaDB, a vector database.
- Performed hybrid search using both keyword matching and semantic similarity.
- Enriched each article with relevant media content (images/videos).
- Delivered top-ranked results via a Telegram bot with summarization.
- Allowed real-time configuration (e.g., sources, timeframe) via Google Sheets.

TELEGRAM COMMANDS:

Command	Description
/start, /help, /settings	Display welcome message and available commands
/news [keyword]	Get latest news about a topic
/semantic [query]	Perform semantic search on recent articles
/news_by_date [YYYY-MM-DD] [keyword]	Get news for a specific date
/news_with_media [keyword]	Get news articles with related media
/media [keyword]	Show media gallery for a keyword
/options, options	View current configuration
/domains	Show news sources used
/timeframe	Change search range in days
/save_to_sheets [keyword]	Save article summaries to Google Sheets
/export_daily	Export today's news for all topics to Google Sheets

HOW INFORMATION RETRIEVAL AND WEB SEARCH IS USED:

1. Keyword-based Retrieval (Classic IR):

Users can input a topic or keyword to retrieve matching articles. This mimics the traditional vector space model in IR, where TF-IDF and keyword frequency are used to find relevant documents.

2. Semantic Search (Modern IR):

Implemented using **SentenceTransformer**, semantic embeddings allow the system to understand the meaning behind a user's query. Instead of just matching keywords, it finds conceptually similar news using vector similarity (cosine distance) between the query and article embeddings.

3. Hybrid Ranking Strategy:

A combination of both **keyword scoring** and **semantic similarity** ensures a balanced ranking. Keyword relevance ensures precision, while semantic matching improves recall, especially for queries with varied phrasing.

4. Content Summarization (Document Processing):

To improve retrieval efficiency, only the relevant portions of news are retained via summarization, a key part of IR preprocessing to reduce document length and redundancy.

5. Real-time Query Handling via Telegram Bot:

Every user command acts as a query, with results fetched, ranked, and returned dynamically – a direct application of **interactive IR** and **query-based filtering**.

6. Structured and Unstructured Data Handling:

Structured data (sources, config) is handled via Google Sheets. Unstructured news articles are processed, summarized, and embedded – showcasing real-world document retrieval challenges.

CONCLUSION:

NewsHive effectively demonstrates the practical application of **Information Retrieval and Web Search** techniques. From keyword-based filtering to advanced semantic search, it combines classical and modern IR approaches to deliver personalized, enriched news experiences. By integrating external APIs, media content, and dynamic user interaction through Telegram, the system offers a seamless and smart way to consume news. Additionally, the use of Google Sheets for real-time control and logging makes it flexible and user-friendly.

This project not only showcases core IR principles but also emphasizes the importance of **intelligent ranking**, **semantic understanding**, and **multi-modal content enrichment** in modern search systems.